

Task Goal g :

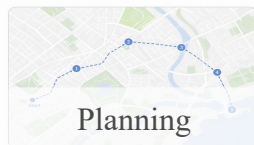
Find the pot and put it on the burner.

Current Context x_t

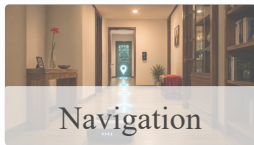
Search in kitchen



World Potential Model (WPM) Φ



Planning



Navigation



Science



Coding



Web

Pretrained World Knowledge

...

Progress-Sensitive Policy Optimization

$$\Delta\Phi = \gamma\Phi_g(x_{t+1}) - \Phi_g(x_t)$$

> 0 ≈ 0 < 0
progress stagnation regression

Progress Signal

Progress comparison

$$x_t \geqslant_g m_0$$

$$x_t \geqslant_g m_1$$

$$x_t \geqslant_g m_2$$

$$m_3 \geqslant_g x_t$$

$$m_4 \geqslant_g x_t$$

$$\Phi_g(x_t) = \phi_2 = 0.5$$

$$\phi_0 = 0.0$$

m_0 : Task start

$$\phi_1 = 0.25$$

m_1 : Hallway reached

$$\phi_2 = 0.5$$

m_2 : Kitchen reached

$$\phi_3 = 0.75$$

m_3 : Pot located

$$\phi_4 = 1.0$$

m_4 : Pot on burner

Generated milestones

